

Unit 4, Lesson 3: Finding the Area of Polygons

Benchmark

MA.7.GR.1.2 Solve mathematical or real-world problems involving the area of polygons or composite figures by decomposing them into triangles or quadrilaterals.

Learning Targets

- ☐ I can decompose a regular polygon into triangles to find the area.
- ☐ I can decompose a regular polygon into triangles and quadrilaterals to find the area.

4.3.1 Warm-Up: Which One?

The dimensions of two rectangles are given.

Rectangle A

13 units by 15 units

Rectangle B

2 units by 3 units

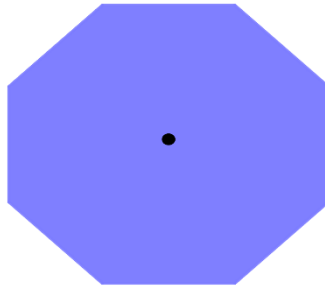
1. Explain which rectangle is more like a square.

4.3.2 Guided Instruction: Decomposing Regular Polygons

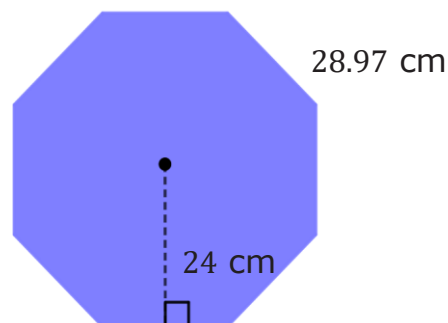


A **regular polygon** is a polygon that is both equilateral (all sides congruent) and equiangular (all angles congruent).

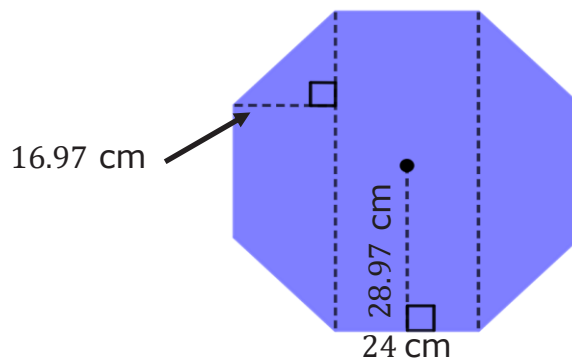
1. A regular octagon is shown.



- On the figure, mark all congruent angles and all congruent sides.
 - Draw lines to decompose the regular octagon into congruent triangles.
 - How many congruent triangles did you decompose the regular octagon into?
 - Discuss with your partner how decomposing the octagon into congruent triangles is helpful in determining the area.
2. Some dimensions of an octagon are given in centimeters (cm). Each side of the octagon has a length of 24 cm. The perpendicular distance from the center to the midpoint of a side is 28.97 cm.



- a. Find the area of one decomposed triangle.
- b. Find the area of the regular octagon to the nearest hundredth of a square centimeter (cm^2) by finding the sum of the areas of all the decomposed triangles.
3. An octagon is divided into two trapezoids and a rectangle. Each side of the octagon has a length of 24 cm. The perpendicular distance from the center to the midpoint of a side is 28.97 cm. The height of each trapezoid is 16.97 cm.



- a. Find the area of each decomposed quadrilateral to the nearest hundredth.
- b. Using the decomposed shapes, determine the area of the octagon to the nearest hundredth of a square centimeter (cm^2).

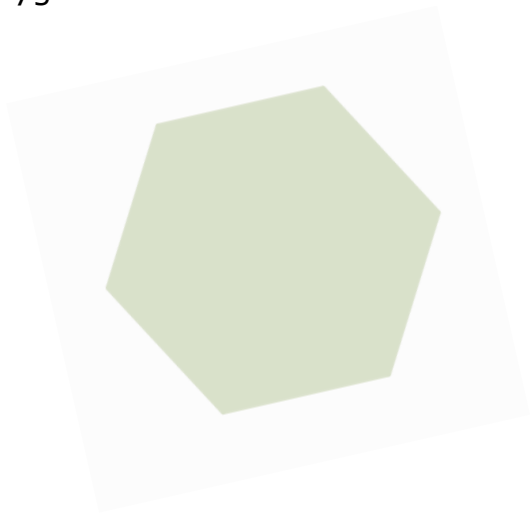
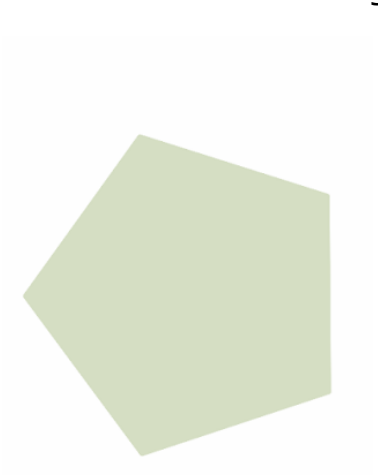
4. Complete the chart using the areas from question 2b and 3b.

Decomposition	Area of Octagon
triangles	
quadrilaterals	

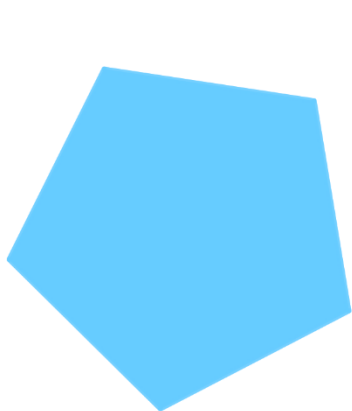
5. What do you notice about the areas?

4.3.3 Collaboration: Decomposing Regular Polygons to Find Area

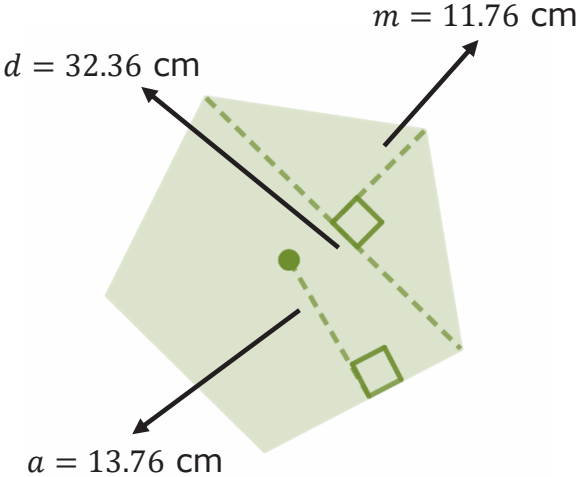
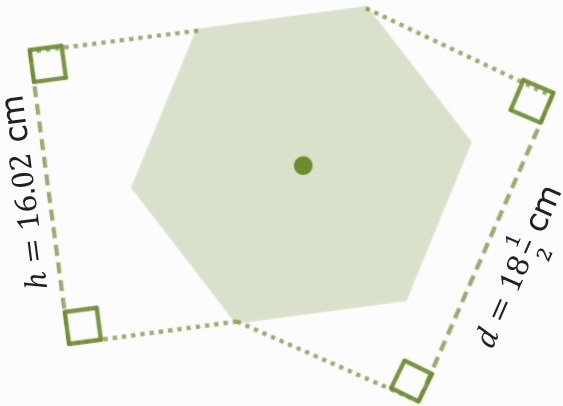
1. Decompose each of the following regular polygons.



2. Share with your partner how you decomposed your shapes. If your partner decomposed the shapes differently, draw on the blue shapes how they decomposed their shapes. If you decomposed the shapes in the same way, work together to create a different way to decompose the shapes.

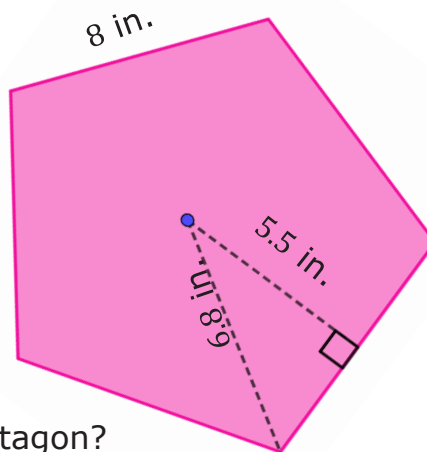


3. The table shows a pentagon and a hexagon. Given measurements in centimeters (cm). Work with your partner to determine the area rounded to the nearest hundredth.

Pentagon	Hexagon
 Side lengths = 20 cm	 Side lengths = $9\frac{1}{4}$ cm
Area of Pentagon	Area of Hexagon

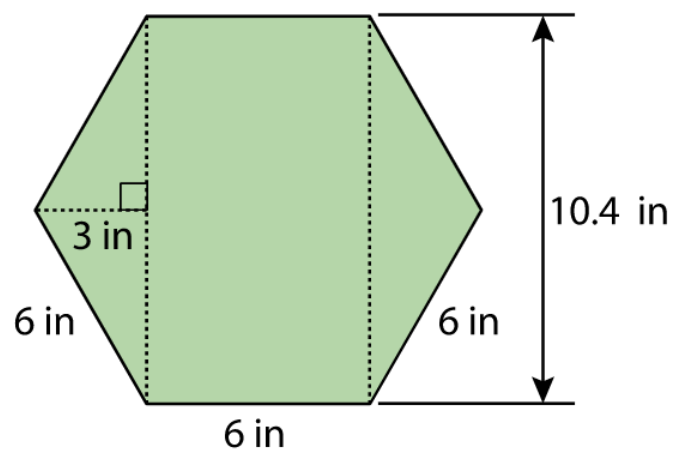
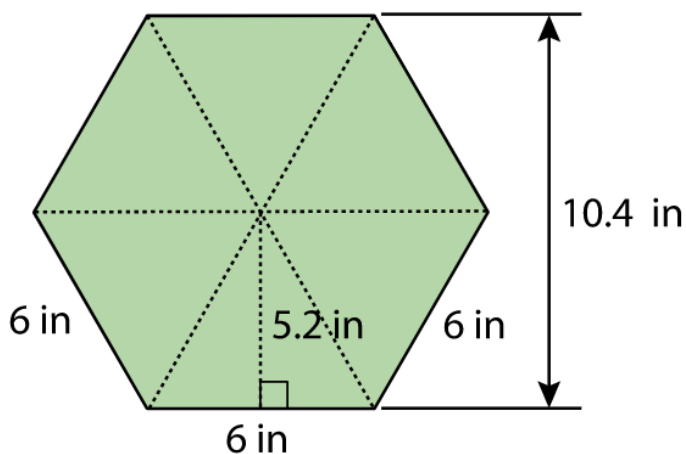
4.3.4 Your Turn!

1. A regular pentagon is shown with lengths in inches (in.).



What is the area of the pentagon?

2. Lin and Andre used different methods to find the area of a regular hexagon with 6 in. sides. Lin decomposed the hexagon into six identical, equilateral triangles. Andre decomposed the hexagon into a rectangle and two triangles.

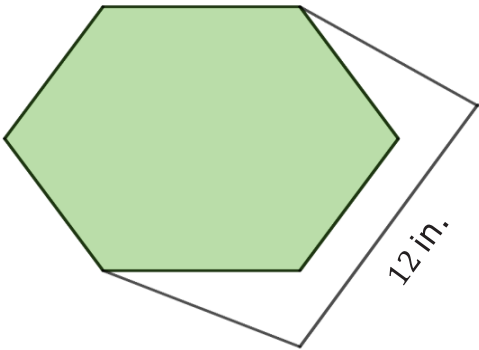


- a. Explain why Lin and Andre will find the same area of the hexagon.

b. Find the area of the hexagon using both Lin’s and Andre’s methods.

Lin’s Method	Andre’s Method

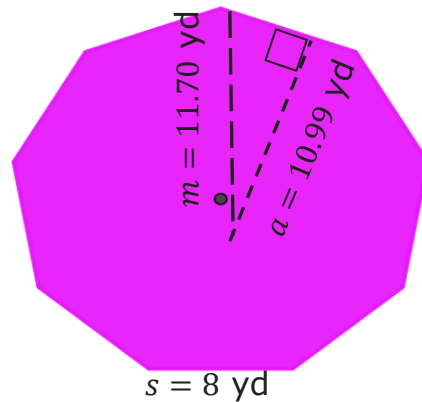
c. Decompose the shape differently than Lin and Andre did.



d. Using the measurements given in all three figures, find the area by finding the sum of your decomposed parts.

4.3.5 Wrap-Up: Error Analysis

1. Vernon found the area of the nonagon. His work is shown.

**Vernon's Work**

Area of Triangle

$$A = \frac{1}{2}(8)(11.70)$$
$$A = 46.8 \text{ yd}^2$$

Area of Nonagon

$$A = 18(46.8)$$
$$A = 842.4 \text{ yd}^2$$

- a. Explain to Vernon what error(s) he made in his work.
- b. What is the area of the nonagon?